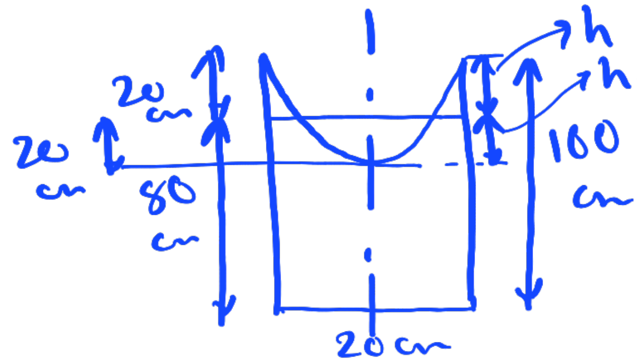
An open circular cylinder of 20 cm diameter and 100 cm long contains water upto a height of 80 cm. It is rotated about its vertical axis. Find the speed of rotation when :
(*i*) no water spills, (*ii*) axial depth is zero. [**Ans.** (*i*) 267.51 r.p.m., (*ii*) 422.98 r.p.m.]



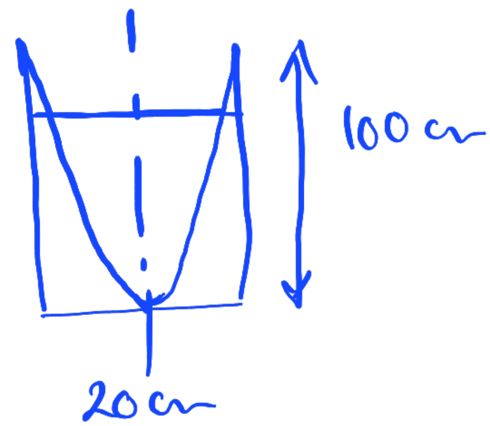
i)

$$Z = \frac{r^2 w^2}{2g}$$

$$\Rightarrow 0.4 = \frac{0.1^2 \times w^2}{2g}$$

$$\Rightarrow w = 28.01 \text{ rad/s}$$

$$= 267.48 \text{ rpm}$$



ii

$$z = \frac{r^2 w^2}{2g}$$

$$\Rightarrow 1 = \frac{0.1^2 \times w^2}{2 \times 9.81}$$

$$\Rightarrow w = 44.29 \text{ rad/s}$$

$$= 422.92 \text{ rpm}$$

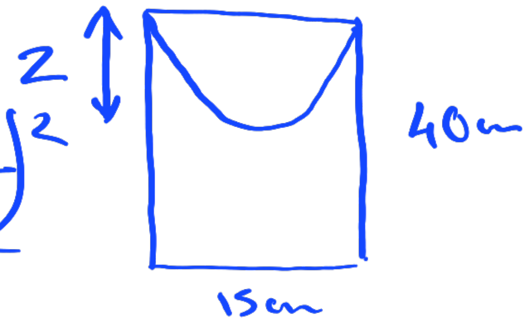
$$\frac{\pi r^2 h}{2}$$

A cylindrical vessel 15 cm in diameter and 40 cm long is completely filled with water. The vessel is open at the top. Find the quantity of water left in the vessel, when it is rotated about its vertical axis with a speed of 300 r.p.m. [**Ans.** 4566.3 cm$^{\cancel{2}\,3}$]

$$Z = \frac{r^2 w^2}{2g}$$

$$= \frac{\left(\frac{15}{200}\right)^2 \times \left(\frac{300 \times 2\pi}{60}\right)^2}{2 \times 9.81}$$

$$= 0.283\,m$$



Vol of cylinder $= \frac{\pi}{4} \times 15^2 \times 40$

$= 7069.5\ cm^3$

Vol of paraboloid $= \frac{\pi \times 7.5^2 \times 28.3}{2} = 2500.8\ cm^3$

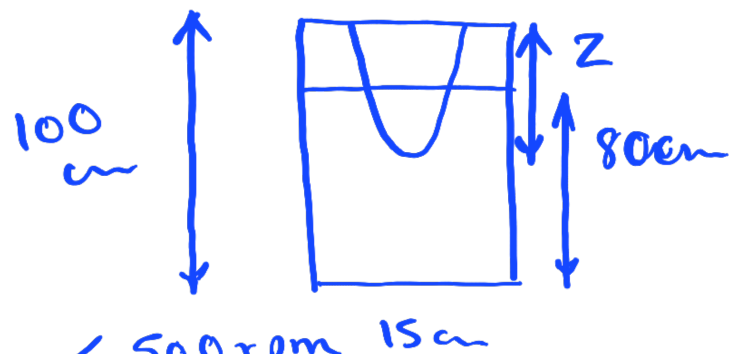
Vol of water gone $= 7069.5 - 2500.8$

$= 4568.6\ cm^3$

A closed cylindrical vessel of diameter 15 cm and length 100 cm contains water upto a height of 80 cm. The vessel is rotated at a speed of 500 r.p.m. about its vertical axis. Find the height of paraboloid formed. [**Ans.** 56.06 cm]

$$z = \frac{r^2 w^2}{2g}$$

$$\Rightarrow 0.4 = \frac{\left(\frac{0.15}{2}\right)^2 \times w^2}{2 \times 9.81}$$



$$\Rightarrow w = 356.6 \text{ rpm} < 500 \text{ rpm}$$

$$z = \frac{r^2 w^2}{2g} \Rightarrow z = \frac{r^2 \times \left(\frac{500 \times 2\pi}{60}\right)^2}{2 \times 9.81}$$

$$z = r^2 \times 139.76 \quad \text{— (1)}$$

$$\frac{\pi r^2 z}{2} = \frac{\pi}{4} \times 0.15^2 \times 0.2 \quad \text{— (2)}$$

vol of paraboloid = vol of air initially present.

from (1) and (2)

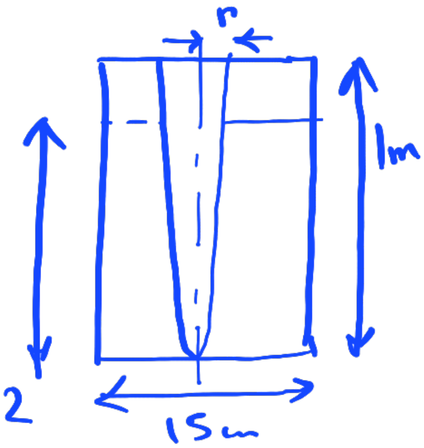
$$z = 56.06 \text{ cm}$$

For the data given in question 20, find the speed of rotation of the vessel, when axial depth is zero.

[**Ans.** 891.7 r.p.m.]

$$z = \frac{r^2 w^2}{2g}$$

$$\Rightarrow 1 = \frac{r^2 w^2}{2g} \quad \text{--- (1)}$$



$$\frac{\cancel{\pi} \times r^2 \times 1}{\cancel{2}} = \frac{\cancel{\pi}}{\cancel{4}\,2} \times 0.15^2 \times 0.2$$

$$\Rightarrow r^2 = 0.00225\ m^2$$

$$1 = \frac{r^2 w^2}{2g}$$

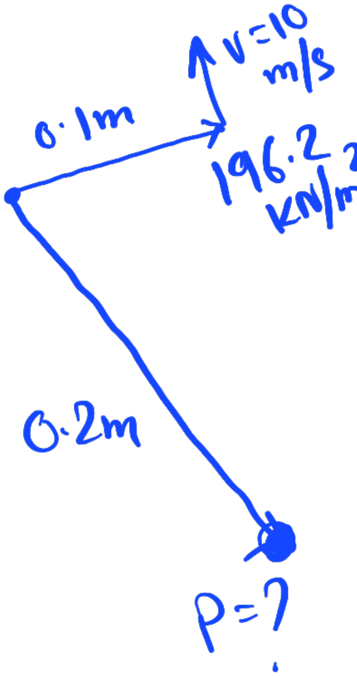
$$\Rightarrow w^2 = \frac{2g}{r^2} = \frac{2g}{0.00225}$$

$$\Rightarrow w = 93.38\ rad/s$$

$$= 891.60\ rpm$$

In a free cylindrical vortex of water, the tangential velocity at a radius of 0.1 m from the axis of rotation is found to be 10 m/s and the intensity of pressure is 196.2 kN/m² [2 kg (f) / cm²]. Find the intensity of pressure at a radius of 0.2 m from the axis.

[*Ans.* 233.7 kN/m², or 233.7 kPa, or 2.382 kg(f)/cm²]



$$v_1 r_1 = v_2 r_2$$

$$\Rightarrow 10 \times 0.1 = v_2 \times 0.2$$

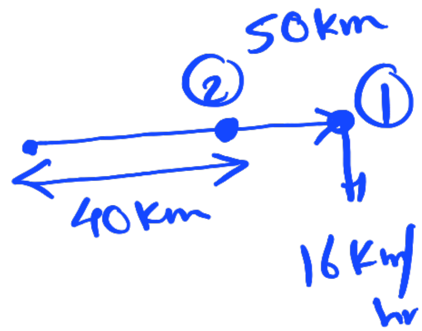
$$\Rightarrow v_2 = 5 \text{ m/s}$$

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + \cancel{z_1} = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + \cancel{z_2}$$

$$\Rightarrow \frac{196.2}{9.81} + \frac{10^2}{2 \times 9.81} = \frac{P_2}{9.81} + \frac{5^2}{2 \times 9.81}$$

$$\Rightarrow P_2 = 233.7 \text{ KPa}$$

The wind velocity in a cyclone may be assumed to be free vortex flow. If at 50 km from centre of cyclone, velocity is 16 km/h, what pressure gradient should be obtained at this point. What reduction in barometric pressure should occur over radial distance of 10 km from this point towards centre of storm. Mass density of air = 1.2 kg/m³. [*Ans.* 1 in 2110; 0.566 m]



$$\frac{\partial P}{\partial r} = \frac{\rho v^2}{r} = \frac{1.2 \times \left(\frac{16 \times 1000}{60 \times 60}\right)^2}{50 \times 1000}$$

$$= 0.000474$$

$$\approx 1 : 2110$$

$$V_1 r_1 = v_2 r_2$$

$$\Rightarrow \quad 16 \times 50 = V_2 \times 40$$

$$\Rightarrow v_2 = 20 \text{ km/hr.}$$

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + \cancel{z_1} = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + \cancel{z_2}$$

$$\Rightarrow \frac{P_1 - P_2}{\rho g} = \frac{v_2^2}{2g} - \frac{v_1^2}{2g}$$

$$= \frac{\left(\frac{20 \times 1000}{60 \times 60}\right)^2 - \left(\frac{16 \times 1000}{60 \times 60}\right)^2}{2 \times 9.81}$$

Pressure diff in terms of head of air = 0.566 m of air that has density of 1.2 kg/m^3

Pressure diff in Pa = 0.566 m × 1.2 × 9.81

= 6.66 Pa